



CCC PLC – Technical Requirements Document

DIGITAL FILING CABINATE (DFC)

Digital Filing Cabinet for CCC PLC – Technical Requirements Document

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Executive Summary

To manage its growing volume of documents, CCC PLC needs a digital filing cabinet that offers efficient storage, preservation, retrieval, and advanced security for its digital records.

The Digital Filing Cabinet (DFC) is a secure, scalable, and compliant document management system designed for institutions. It enables efficient storage, retrieval, classification, and sharing of sensitive financial documents across departments such as Engagements, Accounting, IT, Compliance, Risk, and Audit. The system supports regulatory mandates, internal workflows, and customer service operations while ensuring data integrity and confidentiality.

Scope and Purpose

Purpose

A digital filing cabinet in a financial institution serves as a secure, efficient, and scalable solution for managing vast volumes of sensitive and regulatory documents. Its core purposes include:

- **Document Organization:** Systematically store financial records, client files, compliance documents, and operational paperwork in digital formats.
- **Regulatory Compliance:** Ensure adherence to financial regulations (e.g., KYC, AML, GDPR) through structured recordkeeping and audit trails.
- **Data Security:** Protect confidential information with encryption, access controls, and secure authentication protocols.
- **Operational Efficiency:** Enable quick retrieval, sharing, and processing of documents across departments and branches.
- **Disaster Recovery:** Safeguard critical data through automated backups and cloud-based redundancy.
- **Paperless Transformation:** Reduce reliance on physical storage, promoting sustainability and cost savings.
- **Categorizing Files:** Using file cabinets to categorize data provides a structured and reliable method for organizing physical and digital records, enabling efficient retrieval and consistent indexing.
- By establishing clear labeling conventions, hierarchical folder systems, and standardized metadata, organizations can reduce redundancy, minimize retrieval time, and maintain data integrity.

Scope

The digital filing cabinet supports a wide range of functions within the financial institution, including:

- **Client Services:** Store onboarding documents, account applications, loan agreements, and correspondence.
- **Compliance and Risk Management:** Archive regulatory filings, audit reports, and risk assessments.
- **Finance and Accounting:** Manage invoices, receipts, tax records, and financial statements.
- **Human Resources:** Maintain employee records, contracts, and performance reviews.
- **Executive and Legal:** Secure board minutes, legal agreements, and strategic documentation.

CORE FEATURES OF CCC PLC'S DFC (DIGITAL FILING CABINET)

Document Management

- **Flexible Folder Hierarchy:** Create and customize folder structures to match organizational workflows.
- **Tagging & Metadata:** Apply descriptive tags and attributes for advanced categorization and faster retrieval.
- **Version Control:** Maintain a complete history of document revisions with the ability to restore prior versions.

Security & Access Control

- **User Authentication:** Secure login with password protection and multi-factor authentication (MFA).
- **Role-Based Access Control (RBAC):** Granular permissions assigned by user role, department, or project.
- **Encryption:** End-to-end encryption for data at rest and in transit using industry-standard protocols (AES-256, TLS 1.3).
- **Audit Trails:** Comprehensive activity logs tracking every access, modification, or deletion for full traceability and compliance.

Search & Retrieval

- **Full-Text Search:** Instantly locate documents by keywords or phrases within file content.
- **Advanced Filtering:** Narrow results by date, file type, tags, author, or department.
- **Optical Character Recognition (OCR):** Automatically extract and index text from scanned or image-based documents to enable content search.

Storage & Integration

- **Local Storage:** Securely retrieve and manage files within CCC Plc's intranet.
- **Elastic Scalability:** Seamlessly expand storage capacity without hardware limitations.

Compliance & Reporting

- **Retention Policies:** Automatically archive or delete files based on predefined schedules and regulatory requirements.
- **Analytics & Reporting:** Generate comprehensive reports on access patterns, usage statistics, and audit events.

User Experience

- **Intuitive Interface:** Modern design with drag-and-drop upload, contextual menus, and quick actions.
- **Mobile Accessibility:** Fully responsive interface optimized for smartphones and tablets.

FUNCTIONAL REQUIREMENTS

Folder Structure & Organization

1. Overview

The Digital Filing Cabinet (DFC) shall provide a structured, hierarchical folder system for organizing, storing, and managing digital documents. The folder management module shall enable users to create, maintain, and navigate folders efficiently while enforcing security, access control, and compliance requirements.

2. Functional Requirements for Document Management

2.1. Folder Hierarchy

1. The system **shall** support a hierarchical folder structure with unlimited levels of subfolders.
 2. The system **shall** allow administrators to create a **root folder** for each tenant or department.
 3. The system **shall** enable users (with proper permissions) to create, rename, move, and delete folders.
 4. The system **shall** maintain the logical relationship between parent and child folders.
 5. The system **shall** update the folder path dynamically when folders are renamed or moved.
- The system **shall** preserve document references and metadata during folder moves or reorganizations.

2.2. Standard folder patterns and examples

- Customer Records / [CustomerID] / Profile / Identification

- Accounts and Transactions / [AccountNumber] / Statements / YYYY
- Loans and Credit / [LoanID] / Application / IncomeProof

2.3. File naming convention

- Use a single-line, underscore-delimited format for consistency and sorting: YYYY-MM-DD_CustomerID_DocType_ShortDesc_V{n} Example: 2025-10-20_123456_Passport_Scan_V1

2.4. Folder / Document Properties

- Each folder /document **shall** have a unique system-generated identifier (UUID).
- Each folder/document **shall** include the following metadata fields:
 1. Folder/Document Name
 2. Parent Folder ID
 3. Path
 4. Created By / Modified By
 5. Date Created / Date Modified
 6. Access Permissions
 7. Optional Tags or Categories
 8. Confidentiality Level (Public, Internal, Confidential, Restricted)
 9. Optional Retention Policy ID
- The system **shall** store and update folder metadata in real time.
- The system **shall** display folder properties within the user interface and make them accessible through APIs.

2.5. Access Control & Permissions

- The system **shall** enforce **role-based access control (RBAC)** for folders.
- The system **shall** allow administrators to define permissions such as read, write, share, and delete at both folder and document levels.
- The system **shall** allow permissions to be **inherited** by subfolders and documents from their parent folder.
- The system **shall** allow explicit permission overrides at any level of the hierarchy.

- The system **shall** allow administrators to assign ownership of folders to specific users or roles.
- The system **shall** restrict visibility of sensitive folders (e.g., HR, Legal) based on role or department.

2.6. Folder Management Operations

17. The system **shall** allow authorized users to perform the following actions:

- Create new folders
- Rename existing folders
- Move folders within the hierarchy
- Delete folders (subject to retention and legal hold policies)
- The system **shall** prevent deletion of folders that contain documents under active legal hold or retention.
- The system **shall** allow drag-and-drop reorganization of folders in the user interface.
- The system **shall** support **bulk operations**, allowing multiple folders or documents to be moved or copied simultaneously.
- The system **shall** provide **confirmation prompts** before destructive actions such as delete or move.

2.7. Folder Templates

- The system **shall** allow administrators to define and save **folder templates** for recurring use cases (e.g., Project Folders, Employee Files).
- The system **shall** allow users to instantiate new folder structures from predefined templates.
- The system **shall** allow editing and versioning of folder templates by authorized users.

2.8. Metadata & Tagging

- The system **shall** allow folders to be tagged with user-defined keywords or categories.
- The system **shall** allow metadata and tags to be indexed for fast search and retrieval.
- The system **shall** allow filtering of folders based on metadata, tags, or creation date.
- The system **shall** ensure metadata consistency across all documents within a folder when applied in bulk.

Sample meta data that could be used to create documents:

1. **Title**
2. **Document Type** (controlled list)
3. **Identifier** (Customer ID, Contract Number, Invoice Number)
4. **Date** (YYYY-MM-DD)
5. **Creator** or Source
6. **Department or Owner**
7. **Confidentiality Level** (Public; Internal; Confidential; Restricted)
8. **Retention Period**
9. **Keywords or Tags** Require Document Type, Identifier or Owner, Date, and Confidentiality Level for every archived item.

3.0. FUNCTIONAL REQUIREMENTS FOR Search & Navigation

- The system **shall** support **breadcrumb navigation** to display folder hierarchy paths.
- The system **shall** allow users to search for folders by name, category, metadata, or tags.
- The system **shall** support **smart folders** that display dynamic results based on saved search criteria.
- The system **shall** display search results grouped by category or tag when applicable.
- The system **shall** ensure search results only display documents the user has permission to access.

3.1. Integration & Automation

- The system **shall** integrate folder structures with automated workflows.
- The system **shall** allow documents to be automatically routed to specific folders based on metadata or predefined business rules.
- The system **shall** allow workflow completion (e.g., approval or sign-off) to trigger automatic relocation of documents to designated folders.

4.0. FUNCTIONAL REQUIREMENTS FOR CATEGORIZING OF DOCUMENTS

4.1 Overview

The Digital Filing Cabinet (DFC) shall provide a robust file categorization system that allows users and administrators to organize, classify, and retrieve documents efficiently. Categorization shall be based on metadata, tags, document types, and user-defined attributes to ensure that documents can be grouped, filtered, and located according to business context and compliance requirements.

4.2 Classification Framework

- The system **shall** support multiple methods of file categorization, including:

1. Document Type (e.g., Invoice, Contract, Report, KYC Record)
 2. Department or Business Unit (e.g., HR, Finance, Legal)
 3. Confidentiality Status
 4. Project or Case Reference (e.g., Project ID, Account Number)
 5. User-defined Tags and Metadata Fields.
- The system **shall** allow administrators to define and manage a **taxonomy** or **category hierarchy** to standardize file classification.
 - The system **shall** allow administrators to enforce mandatory metadata fields (e.g., document type, department) for certain categories.
 - The system **shall** allow files to belong to **multiple categories simultaneously** (many-to-many classification).
 - The system **shall** ensure category definitions are consistent and reusable across the organization or tenant.

4.3. Metadata-Based Categorization

- The system **shall** allow users to assign **metadata attributes** to files at the time of upload or afterward.
- The system **shall** support common metadata types, including text, date, numeric, boolean, and list values.
- The system **shall** validate metadata fields according to configured data types and mandatory requirements.
- The system **shall** automatically capture system metadata such as:
 1. Upload date and time
 2. File owner
 3. File type and size
 4. Department or tenant ID
- The system **shall** index all metadata fields for fast search and filtering.

4.4. Tagging System

- The system **shall** allow users to add one or more **tags** (keywords or phrases) to categorize files dynamically.

- The system **shall** allow administrators to define and manage a **controlled vocabulary** or **tag list** to ensure consistency.
- The system **shall** allow users with permission to create **custom tags** when necessary.
- The system **shall** support **auto-suggestions** for tags based on existing tag data.
- The system **shall** allow tags to be color-coded or grouped by category (e.g., “Compliance,” “Pending Review,” “Confidential”).
- The system **shall** store all tag associations in a searchable index.
- The system **shall** prevent duplicate or conflicting tag entries for the same file.

4.5. Automated Categorization

- The system **shall** support **automatic categorization** of files using metadata.
- The system **shall** extract key terms (e.g., customer names, reference numbers, document types) to recommend categories or tags.
- The system **shall** allow administrators to configure **categorization rules** (e.g., “If document contains ‘invoice’, categorize as Finance → Invoices”).
- The system **shall** log all automatic classification actions in the audit log for traceability.
- The system **shall** allow users to manually review and override automatically assigned categories.

4.6. Folder and Category Relationship

- The system **shall** allow documents to be categorized independently of their folder location.
- The system **shall** allow users to view documents by **category, tag, or folder**.
- The system **shall** ensure that moving a document between folders does not remove or alter its category or metadata.
- The system **shall** allow cross-folder and cross-departmental category views (e.g., “All Contracts Across Departments”).

4.7. Confidentiality Classification

1. The system **shall** support multiple confidentiality levels, such as:
 - **Public:** Accessible to all users within the organization.
 - **Internal:** Accessible to authorized employees only.
 - **Confidential:** Restricted to specific departments or roles.
 - **Highly Confidential:** Restricted to designated individuals (e.g., executives, legal).
2. The system **shall** allow administrators to define and customize confidentiality levels as per organizational policy.

3. The system **shall** allow confidentiality classification to be applied automatically or manually during document upload.
4. The system **shall** require users to specify a confidentiality level before uploading or sharing sensitive documents.
5. The system **shall** display a visible **confidentiality label** (e.g., banner, watermark, or icon) on classified documents.
6. The system **shall** enforce access rules based on the assigned confidentiality level.

5.0. FUNCTIONAL REQUIREMENTS FOR Security & Audit & Compliance

- The system **shall** log all folder-related actions (create, rename, move, delete, permission change) in an immutable audit log.
- The system **shall** record audit details including actor ID, action type, timestamp, folder ID, and outcome.
- The system **shall** encrypt all folder metadata and identifiers during transmission and at rest.
- The system **shall** enforce access controls for viewing, editing, or deleting categories and tags.
- The system **shall** restrict modification of system-defined categories to authorized administrators only.
- The system **shall** ensure categorization data (metadata and tags) is encrypted at rest and in transit.
- The system **shall** maintain an immutable **audit trail** of all categorization changes, including:
 1. User ID
 2. Action (add, update, delete)
 3. Timestamp
 4. Previous and new values.

6.0. FUNCTIONAL REQUIREMENTS FOR Performance & Scalability & Storage

- The system **shall** support large-scale folder hierarchies with **at least 10,000 folders per tenant** without performance degradation.
- The system **shall** maintain search and navigation performance with response times under **1 second** for standard queries.

- The system **shall** support parallel access and updates from multiple users without data corruption or lock conflicts.

User Interface Specification — Digital Filing Cabinet (DFC)

1. Overview

The User Interface (UI) of the Digital Filing Cabinet (DFC) shall provide a clean, intuitive, and secure experience for users to upload, organize, retrieve, and manage digital documents.

It shall balance **usability**, **functionality**, and **access control**, ensuring that both technical and non-technical users can interact efficiently with the system across desktop and mobile platforms.

2. Design Objectives

The UI shall:

- Provide an intuitive and consistent layout for all document management tasks.
- Minimize user actions required to perform frequent tasks (upload, search, share, etc.).
- Support accessibility and responsiveness across browsers and devices.
- Reflect confidentiality and access permissions visually.
- Offer a modern, clean aesthetic aligned with corporate design standards.

3. General Layout

The interface shall follow a **three-panel layout** for desktop and a **collapsible panel layout** for mobile:

Navigation Panel (Left):

1. Displays folder hierarchy and user shortcuts.
2. Includes expandable and collapsible folders.
3. Provides quick filters (e.g., Recent Files, Shared With Me, Favorites, Trash).

Content Panel (Center):

1. Displays files and folders within the selected directory.
2. Supports grid and list view modes.
3. Allows sorting by name, size, date modified, and type.
4. Displays icons, confidentiality labels, and version badges.

Details & Actions Panel (Right):

1. Displays selected file's metadata, tags, permissions, and activity log.
2. Contains buttons for preview, download, share, edit metadata, or delete.

4. Primary Interface Components

4.1 Dashboard

- The dashboard shall serve as the user's landing page after login.
- It shall display:
 1. **Recent Files** accessed or modified.
 2. **Pending Approvals** and assigned workflows.
 3. **Storage Usage** overview (used vs. allocated).
 4. **Quick Upload** and **Create Folder** shortcuts.
- The dashboard shall display contextually relevant widgets based on user role (e.g., Compliance Officers see retention alerts).

4.2 Navigation Panel

1. The panel shall show:
 1. Folder tree view (expandable by department or project).
 2. Smart folders (e.g., "All Contracts", "In Review", "Archived").
 3. Favorites and shortcuts.
2. Folder icons shall change visually to reflect:
 1. Locked folders (restricted access)

4.3 File List / Content Area

1. Files shall be displayed in either:
 1. **Grid View:** with thumbnails and document type icons.
 2. **List View:** with detailed columns (Name, Type, Owner, Modified Date, Tags, Confidentiality Level).
2. Each file entry shall include:
 1. Checkbox for selection.
 2. Icon or preview thumbnail.
 3. Name (clickable for preview).
 4. Metadata indicators (e.g., "Version 3", "Confidential").
3. The system shall allow **multi-select** actions (download, move, delete, categorize).
4. Hovering over a file shall display quick-action icons:
 1. Preview
 2. Share

3. Edit Metadata
4. Delete

4.4 Toolbar / Action Bar

- Fixed at the top of the content panel.
- Contains:
 - **Upload** (with dropdown for file or folder upload).
 - **New Folder**
 - **Search Bar**
 - **View Toggle (List/Grid)**
 - **Sort** (by Name, Type, Date, Size)
 - **Filter** (by Tag, Metadata, Owner, Confidentiality)
- Shall update dynamically based on user selection (e.g., shows “Move” or “Share” when items are selected).

4.5 Search & Filter Interface

1. The search bar shall support:
 - Keyword and full-text search.
 - Advanced filter icon that opens a modal for filtering by:
 - Date range
 - File type
 - Tag
 - Owner
 - Confidentiality level
2. Search results shall appear in the content panel with highlighted keywords.
3. The search results page shall allow sorting and export of results.

4.6 Document Preview Panel

1. The system shall allow **in-browser preview** of common file types (PDF, Word, Excel, Images, Text).
2. The preview window shall display:

1. File content (read-only)
 2. Metadata and version info
 3. Tags and comments section
 4. Confidentiality label watermark (if applicable)
3. Users shall be able to:
 1. Add annotations or comments (if permitted).
 2. Toggle between document versions.
 3. Print or download (subject to permissions).

4.7 Metadata & Tag Management

1. The right-side details panel shall display:
 1. Document name, type, size, and modification details.
 2. Metadata fields (e.g., Department, Project, Category).
 3. Tags with color-coded chips.
 4. Confidentiality classification label.
2. Users with edit rights shall be able to:
 1. Add, edit, or remove tags.
 2. Update metadata fields.
 3. Change retention category (if authorized).

4.8 Upload & Drag-and-Drop

1. The interface shall support **drag-and-drop file upload** into any folder.
2. During upload, the system shall display:
 1. Progress bar
 2. Upload status (success, failed, pending)
 3. Auto-assigned metadata (owner, date, file type)
3. Users shall be prompted to enter mandatory metadata (e.g., Document Type, Confidentiality Level) before upload completion.

4.9 Notifications & Alerts

- Notifications shall appear in a top-right dropdown menu or bell icon.

- Types of alerts:
 1. Workflow assignments and approvals
 2. Access requests
 3. Upload completion
 4. Retention expiry or legal hold alerts
- Users shall be able to configure notification preferences (email, in-app, mobile push).

4.10 Sharing & Permissions Modal

1. When a user selects **Share**, a modal shall appear with options to:
 1. Add users or groups by name/email.
 2. Set permissions (View, Edit, Share, Download).
 3. Define expiry date for shared access.
 4. Enable restricted access.
2. The modal shall display a summary of current access permissions.

4.11 Accessibility & Responsiveness

- Support keyboard navigation for all primary functions.
- Fully responsive on:
 - Desktop browsers (Chrome, Edge, Firefox, Safari)
 - Tablets
 - Mobile devices (iOS, Android)

5. Visual Design Standards

- **Color Palette:** Neutral background with accent colors for action buttons and alerts.
- **Typography:** Sans-serif font (e.g., Roboto or Open Sans) for clarity.
- **Icons:** Vector-based (SVG) and consistent with Material Design principles.
- **Buttons:** Rounded corners, hover effects, and consistent sizing.
- **Confidentiality Indicators:**

- *Public* → Gray
- *Internal* → Blue
- *Confidential* → Orange
- *Highly Confidential* → Red

7. Acceptance Criteria

- All primary user actions (upload, search, share, download, delete) are accessible in ≤ 3 clicks.
- Confidentiality labels are visible on all relevant documents.
- UI elements adapt correctly to mobile and tablet form factors.
- Keyboard shortcuts and accessibility features function as intended.
- The interface meets performance, usability, and accessibility standards as tested.

7.0. Non-Functional Requirements

- **Usability:** Folder navigation and management shall be intuitive and consistent across web and mobile interfaces.
- **Reliability:** Folder operations shall maintain data integrity even during concurrent modifications.
- **Auditability:** All actions shall be fully traceable for compliance and review.
- **Extensibility:** The folder module shall support future expansion (e.g., tagging, AI-driven organization, or auto-classification).

8.0. Developer Tasks & Milestones (Suggested Roadmap)

- **Phase 0 — Foundations (4 weeks)** – Software UI Design and Implementation
- **Phase 1 — Ingestion & Storage (6 weeks)** – File Upload Implementation, Integrating 3rd party compression algos.

- **Phase 2 — Search & Classification (6 weeks)** – File indexing and Classification
- **Phase 3 — Security & Compliance (6 weeks)** – Retention policy, encryption, authentication, role-based access
- **Phase 4 — Scale, Monitoring, Hardening (6 weeks)** - Autoscaling, performance tuning, backup policies, Penetration testing & remediation